

7(a)(i)	<table border="1"> <thead> <tr> <th>light intensity / lux</th><th>total CO₂ uptake after 20 seconds / μmol</th><th>rate of photosynthesis / μmols⁻¹</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>0.0</td></tr> <tr> <td>3</td><td>20</td><td>10</td></tr> <tr> <td>6</td><td>44</td><td>22 ;</td></tr> <tr> <td>9</td><td>72</td><td>36</td></tr> <tr> <td>12</td><td>80</td><td>40</td></tr> </tbody> </table>	light intensity / lux	total CO ₂ uptake after 20 seconds / μmol	rate of photosynthesis / μmols ⁻¹	0	0	0.0	3	20	10	6	44	22 ;	9	72	36	12	80	40	1
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7(a)(ii)	five points plotted correctly ; best fit curve drawn for plots ; curve levels off from 12 lux to 18 lux or curve continues up at smaller gradient ;	3																		
7(a)(iii)	<i>curve levels off from 12 lux to 18 lux</i> 1 because light <u>intensity</u> no longer limiting ; 2 temperature / carbon dioxide <u>concentration</u>, now limiting ;	2																		

Question	Answer	Marks
7(b)	any four from: 1 electrons, excited / emitted / described ; 2 electrons travel along ETC and energy released ; 3 (use energy released) to pump H^+ into thylakoid, space/ lumen ; 4 increases concentration of H^+ / protons (in thylakoid space / lumen) / creates proton gradient ; 5 H^+ / protons, diffuse through ATP synthase ; 6 (from thylakoid space) to stroma ; 7 chemiosmosis ;	4